

TechDocRAG: Intelligent Document Parsing and Reasoning System

Project Handbook

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1. Introduction

Problem Background

Organizations process millions of documents daily across formats and types. From financial institutions handling invoices to healthcare providers managing patient records, efficient document understanding has become critical for business operations.

Why Document Parsing is Important

- **Finance:** Invoice processing, contract management, compliance reporting
- **HR:** Resume screening, employee records, skills analysis
- **Legal:** Contract analysis, case preparation, regulatory compliance

- **Healthcare:** Patient records, clinical research, insurance claims

Limitations of Existing Systems

- **OCR-only:** Basic text extraction without context understanding
- **Rule-based:** Rigid, breaks with format changes, high maintenance
- **Vanilla RAG:** Limited numeric reasoning, no multi-document support
- **Current gaps:** No unified solution, lack of explainability, privacy concerns

Our Solution: TechDocRAG

TechDocRAG combines **hybrid extraction + multi-document reasoning + calculations + explainability** through:

- Rule-based methods for structured elements + semantic understanding
 - Cross-document information synthesis
 - Integrated calculation engine for numeric reasoning
 - Full transparency with confidence scores and source attribution
-

2. Problem Statement

Current document parsing systems fail to:

- **Work across different document types** (invoice, resume, contracts, reports)
- **Reason across multiple documents** for comprehensive business insights
- **Perform numeric calculations** (discounts, penalties, due dates)
- **Provide transparency** with explainable answers and confidence scores

Therefore, there is a need for a unified hybrid system that combines rule-based methods and Retrieval-Augmented Generation for reliable, explainable, and domain-independent document intelligence.

3. Objectives

1. Automate structured and unstructured information extraction

- | 95% accuracy for structured fields, >90% for unstructured content

2. Enable multi-document reasoning

- Cross-document validation and information synthesis

3. Support numeric reasoning and calculations

- Business calculations: due dates, penalties, discounts

4. Provide confidence scores + explanations

- Multi-layered confidence scoring with source attribution

5. Ensure privacy by masking sensitive data

- Automated PII detection and on-premises deployment

6. Be adaptable across domains

- Finance, legal, resumes, healthcare with minimal customization
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4. Scope

Included (Phase 1)

- **Document Types:** Invoices, purchase orders, resumes, legal agreements, medical reports
- **Formats:** PDF, DOCX, images with OCR support
- **Features:** Multi-document reasoning, calculations, explainability
- **Languages:** English only

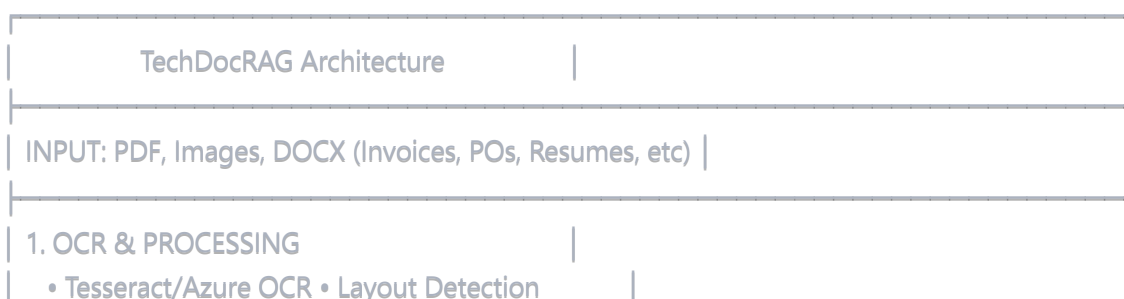
Excluded (Phase 1)

- Handwritten documents, real-time processing, multilingual support
- Enterprise integrations, mobile apps, workflow automation

Future Phases

- **Phase 2:** Multilingual, handwritten docs, mobile apps
 - **Phase 3:** Enterprise integration, advanced analytics
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5. System Architecture





Processing Steps:

- 1. **Document Upload & OCR** → Text extraction with layout understanding
- 2. **Preprocessing & Chunking** → Semantic segmentation with metadata
- 3. **Storage** → Vector embeddings + keyword indexing
- 4. **Query Processing** → Intent analysis and parameter extraction
- 5. **Hybrid Retrieval** → BM25 + embeddings fusion
- 6. **Reasoning** → Rule-based + LLM + calculations
- 7. **Multi-Document Aggregation** → Cross-document synthesis
- 8. **Output** → Explainable answers with confidence scores

6. Key Features

Hybrid Extraction (Rule-based + RAG)

- **Rule-based:** Precise structured field extraction (dates, amounts, IDs)
- **RAG:** Semantic understanding for unstructured queries

- **Integration:** Confidence-weighted combination with intelligent routing

Multi-Document Reasoning

- Automatic relationship detection between related documents
- Cross-document validation and information synthesis
- Timeline construction and process tracking

Automatic Calculations

- Business logic: due dates, penalties, discounts, tax calculations
- Mathematical operations with error handling and validation
- Integration with natural language understanding

Explainable Outputs

- Multi-layered confidence scoring (retrieval, reasoning, calculation)
- Complete source attribution with document references
- Step-by-step reasoning chain visualization

Privacy & Security

- Automated PII detection and masking
 - On-premises deployment options
 - Role-based access control and audit logging
-

7. Tech Stack

Core Components

- **OCR:** Tesseract, AWS Textract, Azure Form Recognizer
- **Embeddings:** Sentence-Transformers (all-MiniLM-L6-v2), OpenAI embeddings
- **Vector DB:** FAISS (dev), Chroma/Pinecone (production)
- **Backend:** Python, FastAPI, PostgreSQL, Redis
- **Frontend:** Streamlit (demo), React (production)
- **LLM:** GPT-4, GPT-3.5-turbo, LLaMA 2 (on-premises)

Infrastructure

- **Containerization:** Docker, Kubernetes

- **Monitoring:** Prometheus, Grafana
 - **Security:** OAuth2, encryption at rest/transit
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8. Datasets

Primary Datasets

- **SROIE:** 1,000+ invoices with ground truth annotations
- **CUAD:** 13,000+ legal contracts with clause annotations
- **MIMIC:** De-identified healthcare records and clinical notes
- **Resume Datasets:** 10,000+ resumes from Kaggle and custom sources

Multi-Document Collections

- Complete procurement cycles (RFP → PO → Invoice)
 - Legal case files with related document chains
 - Patient records with longitudinal medical history
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9. Evaluation Metrics

Accuracy Metrics

- **Field Extraction:** Precision >95%, Recall >93%, F1 >92%
- **Query Answering:** >90% accuracy on benchmark datasets
- **Calculations:** >98% accuracy for mathematical operations
- **Confidence Calibration:** >0.85 correlation with actual accuracy

Performance Metrics

- **Response Time:** <3s single-doc, <10s multi-doc queries
- **Throughput:** 500+ queries/minute, 100+ concurrent users
- **Processing:** <2 minutes per 10-page document

Explainability Metrics

- **Source Attribution:** 100% of answers include references
 - **User Satisfaction:** >4.5/5 rating for explanation quality
-

10. Use Cases (Examples)

Finance

Query: *"If I pay this invoice 12 days late, how much do I owe including penalties?"* **Process:** Extract amount → Calculate due date → Apply penalty rate → Return total **Output:** Original: \$5,000, Penalty: \$65.75, Total: \$5,065.75

Resume/HR

Query: *"List candidates with 3+ years Python experience and CS degree"* **Process:** Parse resumes → Extract skills/education → Filter criteria → Rank results **Output:** Filtered candidate list with anonymized details and confidence scores

Legal

Query: *"What is the penalty clause in this contract and when does it apply?"*
Process: Identify penalty sections → Extract conditions → Calculate amounts → Explain applicability
Output: Detailed penalty terms with contract references and trigger conditions

Healthcare

Query: *"Summarize patient's lab results from last 6 months and flag concerns"* **Process:** Extract lab values → Create timeline → Identify abnormal ranges → Detect trends
Output: Plain English summary with trend analysis and flagged concerns

Multi-Document

Query: *"Match this invoice to corresponding PO and verify amounts"* **Process:** Find related PO → Compare line items → Identify discrepancies → Generate report **Output:** Detailed comparison showing matches, differences, and exceptions

11. Research Gaps Addressed

1. Hybrid Architecture Innovation

Gap: Most systems use either purely neural or purely symbolic approaches **Solution:** Novel integration of rule-based + RAG with intelligent routing **Impact:** First comprehensive hybrid evaluation in document AI

2. Multi-Document Reasoning

Gap: Existing systems process documents in isolation **Solution:** Cross-document synthesis with

relationship detection **Impact:** First effective multi-document reasoning in document AI

3. Integrated Numeric Reasoning

Gap: Document AI cannot perform calculations on extracted data

Solution: Seamless calculation engine integration with NLP **Impact:** First system combining symbolic math with neural text understanding

4. Explainable Document AI

Gap: Black-box solutions without reasoning transparency **Solution:** Multi-layered confidence + reasoning chain visualization

Impact: Comprehensive explainability framework for document processing

5. Domain Adaptability

Gap: Systems require extensive retraining for new domains **Solution:** Modular architecture with transfer learning capabilities **Impact:** Superior cross-domain adaptation with minimal training

6. Privacy-Preserving Intelligence

Gap: Advanced systems require cloud processing **Solution:** Full on-premises capability with automated PII protection **Impact:** First enterprise-grade privacy-compliant document AI

12. Deliverables

Technical Deliverables

- **Working Prototype:** Web app supporting all document types
- **System Architecture:** Complete technical documentation and APIs
- **Core Components:** Processing pipeline, retrieval system, reasoning engine

Research Deliverables

- **Research Papers:** Primary paper on hybrid RAG + secondary papers
- **Benchmarking:** Comprehensive evaluation vs existing systems
- **Open Source:** Core framework components and evaluation tools

Business Deliverables

- **Demo Materials:** Interactive demos and case studies
- **Documentation:** Implementation guides and training materials

- **Compliance:** Security audits and regulatory compliance documentation

Example Outputs

- Structured JSON responses for API integration
 - Human-readable explanations with source references
 - Interactive visualizations for complex relationships
 - Audit trails and compliance reports
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13. Future Scope

Phase 2: Enhanced Capabilities (6-12 months)

- **Multilingual Support:** 10+ major languages
- **Handwritten Documents:** Advanced handwriting recognition
- **Real-time Processing:** <1 second response times
- **Mobile Applications:** iOS and Android apps with offline processing

Phase 3: Enterprise Integration (12-24 months)

- **ERP Integration:** SAP, Oracle, Salesforce connectors
- **Advanced Analytics:** Predictive insights and trend analysis
- **Workflow Automation:** RPA integration and process management
- **Advanced Security:** Quantum-resistant encryption preparation

Long-term Vision (3-5 years)

- **AGI Integration:** Human-level document understanding
 - **Industry Verticals:** Specialized healthcare, finance, legal solutions
 - **Global Scale:** 50+ languages with cultural context understanding
 - **Quantum Computing:** Quantum-enhanced optimization and security
-

Implementation Quick Start

Development Setup

```
bash
```

Prerequisites

Python 3.9+, PostgreSQL, Redis, Docker

Installation

```
git clone https://github.com/your-org/techdocrag
```

```
cd techdocrag
```

```
pip install -r requirements.txt
```

```
docker-compose up -d
```

Configuration

```
export DATABASE_URL=postgresql://user:pass@localhost:5432/techdocrag
```

```
export OPENAI_API_KEY=your_key_here
```

Basic Usage

```
python
```

```
from techdocrag import DocumentProcessor, QueryEngine
```

Process document

```
processor = DocumentProcessor()
```

```
result = processor.process('invoice.pdf', doc_type='invoice')
```

Query system

```
engine = QueryEngine()
```

```
response = engine.query(  
    "What is the total amount due?",  
    doc_ids=[result.doc_id]  
)
```

```
print(f"Answer: {response.answer}")
```

```
print(f"Confidence: {response.confidence}")
```

```
print(f"Sources: {response.sources}")
```

Project Timeline

Months 1-4: Foundation (OCR, basic RAG, core features) **Months 5-8:** Advanced features (multi-doc, calculations, explainability)

Months 9-12: Domain expansion and enterprise features **Months 13-16:** Optimization and research publication

TechDocRAG represents a significant advancement in document AI, combining technical innovation with practical business value and strong research contributions. This hybrid approach addresses critical gaps in current solutions while providing a foundation for future document intelligence capabilities.